



HK IMO Training 2023-2024

Problem Set 19 Suggested Solutions



Problem 73 (Ukraine MO 2017 Fourth Round Grade 11 Problem 2). There are $n \geq 2$ cities in a country. There is a road between every pair of cities, and there is a cost to use each road. Suppose the costs of using these roads are $1, 2, 3, \dots, \frac{n(n-1)}{2}$ respectively in some order.

Two people X and Y wish to travel all the cities in the following manner. X begins the journey at any city. Every time he chooses the road with the smallest cost to a city which he has not visited before, until he has visited all n cities. Y begins the journey at any city (not necessarily the same city as X). Every time he chooses the road with the largest cost to a city which he has not visited before, until he has visited all n cities.

Is it possible to assign the costs to the roads in a certain way such that X and Y can travel all the cities with the same total cost?

Solution. Yes. Denote the cities by $A_1, A_2, \dots, A_n$. The cost to travel between A_i and A_j is denoted by $c_{i,j}$. Suppose we assign the costs such that

$$\begin{aligned} c_{1,2} &< c_{1,3} < c_{1,4} < \dots < c_{1,n} \\ &< c_{2,3} < c_{2,4} < \dots < c_{2,n} \\ &< c_{3,4} < \dots < c_{3,n} \\ &< \dots \\ &< c_{n-1,n}. \end{aligned}$$

If X starts at A_1 , he will travel the cities in the order $A_1, A_2, \dots, A_n$. If Y starts at A_n , he will travel the cities in the order $A_n, A_{n-1}, \dots, A_1$. Therefore, the costs of both X and Y are

$$c_{1,2} + c_{2,3} + \dots + c_{n-1,n}.$$

□

Problem 74 (Ukraine MO 2017 Fourth Round Grade 11 Problem 1). Let a, b, c be given positive real numbers. Find the maximum value of

$$|ax + by - cz| + |ax - by + cz| + |-ax + by + cz|$$

where $x, y, z \in [0, 1]$.

Solution. Answer: $3 \max \{a, b, c\}$.

Let $A = ax, B = by, C = cz$, and let

$$S = |A + B - C| + |A - B + C| + |-A + B + C|.$$

WLOG assume $A \geq B \geq C$. Then we have

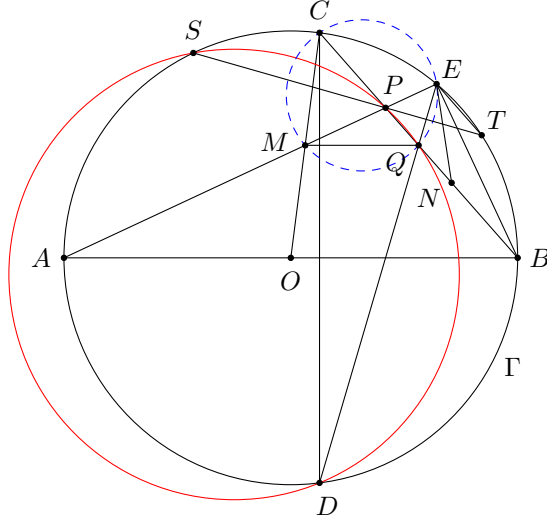
$$S = (A + B - C) + (A - B + C) + |-A + B + C| = 2A + |-A + B + C|.$$

- If $A \geq B + C$, then $S = 2A - (-A + B + C) = 3A - B - C \leq 3a - 0 - 0 = 3a$.
- If $A < B + C$, then $S = 2A + (-A + B + C) = A + B + C \leq A + A + A \leq a + a + a = 3a$.

Thus, we have $S \leq 3a \leq 3 \max \{a, b, c\}$ in any case.

Conversely, WLOG assume $a \geq b, c$. Then $S = 3a = 3 \max \{a, b, c\}$ when $x = 1$ and $y = z = 0$. Therefore, the maximum value of S is $3 \max \{a, b, c\}$. □

Problem 75 (Ukraine MO 2017 Fourth Round Grade 9 Problem 4). Let AB be a diameter of a circle Γ with centre O . A chord CD of Γ is perpendicular to AB . Let M be the midpoint of CO . The line AM meets Γ again at E . The line BC meets AE and DE at P and Q respectively. The circumcircle of $\triangle DPQ$ meets Γ again at S . Prove that $PS = PB$.



Solution. Let T be the second intersection of SP and Γ . Applying Reim's theorem to Γ and (DSP) with lines DQE and SPT , we know that $PQ \parallel ET$. Next, as A is the midpoint of arc CD of Γ , we have

$$\angle MCQ = \angle OCB = \angle OBC = \angle ABC = \angle AED = \angle MEQ.$$

This implies C, M, Q, E are concyclic. Applying Reim's theorem to Γ and (CME) with lines CQB and EMA , we know that $MQ \parallel AB$. As M is the midpoint of CO , Q is the midpoint of CB .

By Menelaus' theorem, we have $\frac{BP}{PC} \times \frac{CM}{MO} \times \frac{OA}{AB} = 1$. This implies $BP = 2PC$. Let N be the midpoint of BP . Then $BN = NP = PC$. Since $\angle PEB = 90^\circ$, $NE = NB = NP$. As $ECBT$ is an isosceles trapezoid, so is $EPNT$. It follows that

$$PC = PN = EN = PT.$$

This implies $PS = PB$ using the power of a point theorem. □

Problem 76 (Brazil Undergraduate MO 2022 Problem 6). Let $p \equiv 3 \pmod{4}$ be a prime number. Suppose θ is an angle such that $\tan \theta \in \mathbb{Q}$. Prove that $\tan((p+1)\theta)$ is a rational number $\frac{a}{b}$ (where $(a, b) = 1$) such that $p \mid a$.

Solution. Let $\tan \theta = \frac{r}{s}$ where $(r, s) = 1$. For each positive integer n , if $\tan n\theta = \frac{r_n}{s_n}$, then

$$\tan((n+1)\theta) = \tan(n\theta + \theta) = \frac{\frac{r_n}{s_n} + \frac{r}{s}}{1 - \frac{r_n}{s_n} \cdot \frac{r}{s}} = \frac{sr_n + rs_n}{ss_n - rr_n}.$$

Therefore, we can define the recurrence relations $r_{n+1} = sr_n + rs_n$ and $s_{n+1} = ss_n - rr_n$ with $r_1 = r$ and $s_1 = s$ such that $\tan n\theta = \frac{r_n}{s_n}$ for all $n \geq 1$. The relations can be rewritten as

$$\begin{pmatrix} r_{n+1} \\ s_{n+1} \end{pmatrix} = \begin{pmatrix} s & r \\ -r & s \end{pmatrix} \begin{pmatrix} r_n \\ s_n \end{pmatrix} = M \begin{pmatrix} r_n \\ s_n \end{pmatrix}.$$

Thus,

$$\begin{pmatrix} r_n \\ s_n \end{pmatrix} = M^{n-1} \begin{pmatrix} r \\ s \end{pmatrix}$$

for all positive integers n . Using the standard diagonalization process (see remark), we obtain

$$M = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix} \begin{pmatrix} s+ri & 0 \\ 0 & s-ri \end{pmatrix} \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix}.$$

This implies

$$\begin{aligned}
M^p &= \frac{1}{2^p} \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix} \begin{pmatrix} (s+ri)^p & 0 \\ 0 & (s-ri)^p \end{pmatrix} \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix} \\
&\equiv \frac{1}{2^p} \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix} \begin{pmatrix} s^p + (ri)^p & 0 \\ 0 & s^p + (-ri)^p \end{pmatrix} \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix} \\
&= \frac{1}{2^p} \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix} \begin{pmatrix} s^p - r^p i & 0 \\ 0 & s^p + r^p i \end{pmatrix} \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix} \\
&\equiv \frac{1}{2} \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix} \begin{pmatrix} s-ri & 0 \\ 0 & s+ri \end{pmatrix} \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix} \\
&= \begin{pmatrix} s & -r \\ r & s \end{pmatrix} \pmod{p}
\end{aligned}$$

by the Fermat little theorem. This implies

$$\begin{pmatrix} r_{p+1} \\ s_{p+1} \end{pmatrix} = M^p \begin{pmatrix} r \\ s \end{pmatrix} \equiv \begin{pmatrix} s & -r \\ r & s \end{pmatrix} \begin{pmatrix} r \\ s \end{pmatrix} \equiv \begin{pmatrix} 0 \\ r^2 + s^2 \end{pmatrix} \pmod{p}.$$

As $p \equiv 3 \pmod{4}$ and $(r, s) = 1$, we have $p \nmid r^2 + s^2$. This shows p divides the numerator of $\tan((p+1)\theta) = \frac{r_{p+1}}{s_{p+1}}$, but does not divide the denominator. $\square$

Remark. Solving $\det(M - tI) = (s-t)^2 + r^2 = 0$ in t , we obtain $t = s \pm ri$. When $t = s + ri$, we solve $(M - (s + ri)I)\mathbf{x} = \mathbf{0}$ to obtain an eigenvector $\mathbf{x} = \begin{pmatrix} 1 \\ i \end{pmatrix}$. When $t = s - ri$, we solve $(M - (s - ri)I)\mathbf{x} = \mathbf{0}$ to obtain an eigenvector $\mathbf{x} = \begin{pmatrix} 1 \\ -i \end{pmatrix}$.

Therefore, we have

$$M = \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix} \begin{pmatrix} s+ri & 0 \\ 0 & s-ri \end{pmatrix} \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix}^{-1} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix} \begin{pmatrix} s+ri & 0 \\ 0 & s-ri \end{pmatrix} \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix}.$$